



Assignment – 1

Design and Analysis of Algorithms (IT2005)
Indian Institute of Information Technology Allahabad

Instructor: K P Singh/Navjot/Naveen

Instructions

- All answers must be clearly justified. Hand written solution is desirable.
- Algorithms must include time and space complexity analysis.
- Use pseudo-code where appropriate.
- Late submissions will not be considered for evaluation.
- Plagiarism will be checked. AI generated submission will be penalized.

Question 1. Consider a building with infinitely many floors. You need to find the highest floor h from which an egg can be dropped without breaking. Can you do it with $O(\log h)$ egg droppings?

Question 2. Given an array with n positive integers $[a_1, a_2, \dots, a_n]$ and a target value S , find the minimum length subarray whose sum is at least S . Can you do it in $O(n \log n)$ time? A subarray means a contiguous subset. That is for some $i \leq j$, $[a_i, a_{i+1}, \dots, a_j]$.

Question 3. Given two sorted integer arrays (with all distinct numbers in them) of size n , we want to find the median of the union of the two arrays. Can you find it by accessing only $O(\log n)$ entries in the two arrays?

Question 4. Given an array of n integers and an integer S , find a pair of integers in the array whose sum is S . Can you do it in $O(n \log n)$ time?

Question 5. Given an integer a , check if it is of the form b^k for some (unknown) integers b and $k > 1$. Can you do this in time $O(\log^c a)$ for some constant c ?

Question 6. You are given a list of landholdings, say $a_1, a_2, \dots, a_n$ (in hectares, can be zero). We want to give land to everyone who has less than f hectares and bring them up to f hectares. For this, we need to take away land from everyone who has more than c hectares, bring them down to c hectares, and redistribute the obtained land.

1. What is the highest value of f that can be feasible?
2. For a chosen value of f , find the right value of c ?

Question 7. You are standing on the number line at $x = 0$. There is a hidden treasure at $x = N$ for some unknown integer N (it could be positive or negative). You have a detector



which will beep if you pass through the location of the treasure. What should be your strategy to minimize the total distance traveled and find the treasure? Can you ensure that the total distance traveled is $O(N)$?

One strategy is to just keep traveling in the positive direction. If the treasure is on the positive side, then you can find it with distance traveled N . But if it is on the negative side, you will never find it. So this strategy does not work.

Question 8. Solve each question using substitution and Guess/Tree Method. Also, solve using Master's Theorem.

1. $T(n) = T(n/2) + n^3$.
2. $T(n) = 4T(n/3) + n$.
3. $T(n) = 4T(n/2) + n$.
4. $T(n) = 3T(n-1) + 1$.
5. $T(n) = 16T(n/4) + n^2$.
6. $T(n) = 4T(n/2) + n^2 \log n$.

Question 9. Suppose there is a trader for a particular commodity, whose license allows them to buy it only once and sell it only once during a season. Naturally, the commodity can be sold only after it is bought. The price of the commodity fluctuates every day. The trader wants to compute the maximum profit they could have made in the last season.

Design an $O(n)$ -time algorithm, where the input is the list of prices $\{p_1, p_2, \dots, p_n\}$ for n days in the last season and the output is the maximum possible profit. Assume addition, subtraction, comparisons, etc., are unit-time operations.

Example input: Prices: 70, 100, 140, 40, 60, 90, 120, 30, 60.

Output: Max profit: 80

Question 10. Celebrity. There is a party with n people, among them there is 1 celebrity. A celebrity is someone who is known to everyone, but she does not know anyone. You need to identify the celebrity by only asking the following kind of queries: ask the i th person if they know the j person. Can you do this in $O(n)$ queries?